

Research Paper

RIT NSBE RESEARCH COMMITTEE
IMAGINE RIT
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PROJECT AURIA: PHOTOVOLTAIC SOLAR PANEL BACKPACK

Research Question : How can current solar backpacks be improved to provide more reliable, eco-friendly charging for students, and communities with limited power access?

Abstract

Reliable access to energy remains a persistent challenge for students and communities that lack consistent electrical infrastructure. Solar-powered backpacks offer a portable and sustainable solution by allowing users to generate energy while moving throughout their day. Despite this potential, many existing designs struggle to provide consistent performance due to limitations in energy efficiency, overheating, durability, and overall usability. These challenges reduce their effectiveness, particularly in environments where dependable charging is essential. Advancements in photovoltaic technology, cooling systems, and smart materials present an opportunity to significantly improve the functionality of these devices. By analyzing current research and engineering developments, this study explores how solar backpacks can be redesigned to optimize energy capture, regulate heat, and better align with user needs. The goal is to develop a more reliable and environmentally sustainable solution that supports both everyday student use and broader applications in communities with limited access to power.

Background/Introduction

The increasing reliance on portable electronic devices has made access to consistent and reliable energy a fundamental part of daily life, particularly for students who depend on phones, laptops, and tablets for academic, professional, and social engagement. Despite this growing dependence, access to stable charging infrastructure is not always guaranteed, especially in outdoor environments, during travel, or within communities where electrical systems are limited or unreliable. Similar to how environmental challenges such as marine plastic pollution in Senegal are intensified by gaps in infrastructure, inefficient systems, and uneven resource distribution, limitations in energy accessibility can also be traced to inefficiencies in design and implementation. Solar-powered backpacks have emerged as an innovative attempt to address this issue by integrating renewable energy into a portable, wearable system, allowing users to generate power throughout the day. However, current models often fall short of their intended purpose due to inconsistent energy output, overheating, limited battery storage, and lack of

user-centered design considerations. These challenges reduce their effectiveness and prevent widespread adoption, particularly in contexts where reliability is essential. At the same time, advancements in photovoltaic technology, energy storage systems, and wearable electronics present a strong opportunity for improvement.

More efficient solar panels can increase energy capture, while improved thermal management strategies can prevent performance loss caused by overheating. Additionally, the integration of lightweight, durable, and potentially adaptive materials can enhance both functionality and comfort, making these systems more practical for everyday use. Addressing these limitations requires a multidisciplinary approach that combines engineering innovation with sustainability principles and human-centered design. By rethinking how solar backpacks are developed and implemented, it becomes possible to create a more reliable, eco-friendly solution that not only supports student needs but also extends to underserved communities facing ongoing challenges with energy access.

Sustainable Development Goals (SDGs) and Their Applications

Improvements in solar backpack technology align with several United Nations Sustainable Development Goals, particularly those centered on energy accessibility, innovation, and sustainable development. Expanding the availability of portable renewable energy contributes directly to SDG 7 by offering a decentralized and sustainable alternative to conventional power sources, especially in regions where electrical infrastructure is insufficient or unreliable. Progress in system design and material efficiency further supports SDG 9 by fostering innovation in wearable technology, energy systems, and product engineering, ultimately strengthening the development of resilient and adaptive infrastructure. Within urban and academic settings, the integration of renewable energy into everyday objects promotes more sustainable consumption patterns, reinforcing the objectives of SDG 11 by reducing dependence on centralized energy systems. Additionally, prioritizing durability, material efficiency, and long-term usability aligns with SDG 12, as it encourages responsible production practices and minimizes environmental impact. Collectively, these connections demonstrate that improving solar backpack technology extends beyond product development, contributing to broader global initiatives focused on sustainability, equity, and technological advancement.

NSBE/ Environmental Solution(s)

Addressing the shortcomings of existing solar backpack systems requires a comprehensive approach that integrates technical innovation with practical design considerations. Enhancing photovoltaic efficiency remains a critical priority, as improved energy conversion enables more reliable performance under varying environmental conditions. The incorporation of lightweight and flexible solar panels can further increase adaptability while maintaining functionality, allowing the system to better accommodate user movement and daily use. Effective thermal

regulation is equally essential, as excessive heat negatively impacts both energy output and component longevity; implementing passive cooling mechanisms, such as optimized airflow structures and thermally conductive materials, can mitigate these effects without introducing unnecessary system complexity. Advancements in energy storage technologies also play a significant role, ensuring that collected energy can be retained and utilized efficiently during periods of limited sunlight. Material selection contributes to both durability and sustainability, with the integration of reinforced or adaptive fabrics enhancing structural integrity and user comfort. Emphasizing accessibility and usability ensures that these technological improvements translate into practical solutions, particularly for students and communities with limited access to reliable power. Through the strategic combination of these elements, solar backpacks can evolve into a more dependable and impactful tool for sustainable energy delivery.

Procedure (4/19/26 will continue)

The development of the Auria solar backpack follows a structured and collaborative design process that integrates research analysis, technical development, and iterative prototyping. Initial stages involve reviewing existing solar backpack technologies and relevant literature to identify limitations in energy efficiency, thermal regulation, and usability, which inform the primary design objectives. Building on these findings, the project is organized into specialized committees, including microcontroller systems, web design and development, and CAD and sewing, each responsible for distinct aspects of the system. The microcontroller committee focuses on programming and circuitry development, requiring the creation of a detailed circuit diagram, a comprehensive list of electronic components, and a clear explanation of the wiring configuration. The system architecture includes a 5V solar panel connected to a TP4056 charging module, which feeds into a battery or power bank, followed by a boost converter that distributes power to both a wireless charging module and a USB port, with additional integration of an ESP32 microcontroller and INA218 power sensor for monitoring. Circuit design and simulation are developed using platforms such as TinkerCAD, KiCad, or Fusion 360.

Concurrently, the web development committee designs a front-end interface that reflects a professional, sustainability-focused aesthetic, with all collaborative work managed and uploaded through GitHub-based platforms. The CAD and sewing committee is responsible for constructing the physical prototype, beginning with a three-dimensional model based on the official design sketch, followed by the integration of structural and electronic components, including flexible solar panels, internal compartments for circuitry, and user-facing features such as a wireless charging pocket and optional GPS module. Prototyping emphasizes functionality, durability, and user comfort, with materials selected to support both performance and environmental considerations. All components and designs are continuously reviewed, uploaded, and refined through shared platforms such as Figma and GitHub to ensure transparency and collaboration. Following assembly, the prototype undergoes testing under varied conditions to evaluate energy output, charging reliability, and thermal performance, with collected data informing subsequent design iterations. This process ensures that the final product reflects both

technical precision and user-centered design, resulting in a practical and sustainable energy solution for students and communities with limited access to reliable power.

Conclusion

The refinement of solar-powered backpack technology represents a significant opportunity to address persistent challenges related to energy accessibility and environmental sustainability. Improvements in efficiency, thermal management, material durability, and overall system integration enhance the practicality and reliability of these devices, allowing them to function effectively in diverse real-world conditions. The incorporation of renewable energy into portable, everyday objects reflects a broader shift toward sustainable innovation, demonstrating how targeted design improvements can contribute to long-term environmental goals. For students, such advancements facilitate consistent access to power, supporting academic engagement and productivity, while for underserved communities, they provide a viable alternative to traditional infrastructure limitations. Continued exploration and development in this field hold the potential to expand access to clean energy solutions while promoting responsible engineering practices, ultimately contributing to a more sustainable and equitable global landscape.

Annotations

1. Abdelmalek, Z., Rehman, K. U., Al-Mdallal, Q. M., Al-Kouz, W., & Malik, M. Y. (2020). Dynamics of thermally magnetized grooved flow field having uniformly heated circular cylinder: Finite element analysis. *Case Studies in Thermal Engineering*, 21. <https://doi.org/10.1016/j.csite.2020.100718>
 - a. Focusing on thermally magnetized fluid flow, Abdelmalek and colleagues analyze how a uniformly heated circular cylinder behaves within a grooved flow field using finite element methods. The research explores how magnetic forces interact with thermal gradients to influence velocity distribution, temperature patterns, and overall heat transfer. Their simulations reveal that both the structure of the flow field and the intensity of magnetic parameters significantly impact system performance, offering insight into complex thermal-fluid interactions. These findings are valuable for engineering applications where heat management and fluid behavior are tightly coupled. In the context of compact electronic systems or solar-powered devices, efficient heat dissipation is critical to maintaining performance. The study provides a theoretical foundation that can help guide the design of systems where airflow and thermal control must be optimized within limited space.
2. Blondeau, J., Museur, T., Demaude, O., Allard, P., Turoni, F., & Mertens, J. (2020). Cost-effective flexibilisation of an 80 MWe retrofitted biomass power plants: Improved combustion control dynamics using virtual air flow sensors. *Case Studies in Thermal Engineering*, 21. <https://doi.org/10.1016/j.csite.2020.100680>
 - a. Improving combustion efficiency in biomass power systems is addressed through the implementation of virtual air flow sensors, which enhance control dynamics without the need for extensive physical instrumentation. The authors demonstrate that data-driven sensor modeling can provide accurate real-time airflow measurements, allowing for more responsive and stable combustion processes. This approach reduces operational costs while maintaining system flexibility, particularly in retrofitted plants where traditional upgrades may be impractical or expensive. The broader significance of this work lies in its emphasis on intelligent system control through computational methods. Although focused on large-scale energy systems, the concept of replacing hardware-intensive monitoring with virtual sensing can be applied to smaller, portable technologies. For energy-efficient device design, integrating software-based monitoring and control mechanisms can improve performance while minimizing complexity, cost, and weight.
3. Budiyo, M. A., & Shinoda, T. (2020). An investigation of thermal stratification in refrigerated container yards. *Case Studies in Thermal Engineering*, 21. <https://doi.org/10.1016/j.csite.2020.100703>
 - a. Thermal stratification within refrigerated container environments is examined to understand how temperature gradients develop and impact cooling efficiency. The study identifies key factors contributing to uneven temperature distribution, including airflow patterns, container arrangement, and external environmental conditions. Findings indicate that inadequate circulation can lead to significant variations in temperature layers, reducing system effectiveness and potentially compromising stored materials. These insights are applicable to the design of any system where internal temperature consistency

is critical. In compact electronic or energy storage devices, uneven heat distribution can lead to inefficiencies or localized overheating. By highlighting the importance of airflow and spatial configuration, this research reinforces the need for thoughtful internal design in systems requiring controlled thermal environments.

4. **Das, M., & Akpinar, E. K. (2020).** Determination of thermal and drying performances of the solar air dryer with solar tracking system: Apple drying test. *Case Studies in Thermal Engineering*, 21. <https://doi.org/10.1016/j.csite.2020.100731>
 - a. Solar energy utilization is explored through the performance evaluation of a solar air dryer equipped with a tracking system. Using apple drying as a case study, the authors demonstrate that solar tracking significantly enhances thermal efficiency by maintaining optimal exposure to sunlight throughout the day. The system achieves higher drying rates and improved energy utilization compared to fixed-position designs, highlighting the importance of dynamic solar alignment. This work underscores the broader principle that maximizing solar energy capture directly improves system performance. In portable solar technologies, such as wearable charging systems, optimizing panel orientation can substantially increase efficiency without requiring additional energy input. The findings provide valuable guidance for designing systems that rely on consistent and effective solar energy harvesting.
5. **Dwivedi, P., Sudhakar, K., Soni, A., Solomin, E., & Kirpichnikova, I. (2020).** Advanced cooling techniques of P.V. modules: A state of art. *Case Studies in Thermal Engineering*, 21, 100674. <https://doi.org/10.1016/J.CSITE.2020.100674>
 - a. A comprehensive synthesis of photovoltaic cooling strategies highlights the critical relationship between operating temperature and energy conversion efficiency. The authors examine a range of cooling approaches, including passive air cooling, active liquid systems, phase change materials, and nanofluid technologies, comparing their effectiveness, cost implications, and scalability. The analysis confirms that elevated temperatures significantly reduce photovoltaic efficiency, reinforcing the necessity of effective thermal management solutions. For small-scale and portable solar applications, the challenge lies in implementing efficient cooling without increasing system complexity or weight. The insights provided in this review support the selection of practical and cost-effective thermal strategies, enabling improved performance in wearable or mobile solar devices. The work serves as a foundational reference for optimizing energy output under real-world operating conditions.
6. **Gong, W., et al. (2022).** Scalable and Reconfigurable Green Electronic Textiles with Personalized Comfort Management. *ACS Nano*, 16(8), 12635–12644. <https://doi.org/10.1021/acsnano.2c04252>
 - a. Advancements in electronic textiles are presented through the development of scalable, flexible materials capable of delivering personalized thermal regulation. The integration of electronic components into fabric enables dynamic temperature control while preserving comfort, flexibility, and mobility. The design emphasizes sustainability and adaptability, allowing the system to be tailored to individual user needs and environmental conditions.

- b. This innovation represents a significant step toward fully integrated wearable energy systems. For applications involving solar-powered backpacks or smart textiles, embedding electronics directly into fabric enhances usability and design efficiency. The study highlights the importance of merging engineering performance with user-centered considerations, supporting the development of multifunctional wearable technologies.
- 7. **Hayat, H. M. A., et al. (2020).** Case studies on the effect of two-dimensional heliostat tracking on the performance of domestic scale solar thermal tower. *Case Studies in Thermal Engineering*, 21. <https://doi.org/10.1016/j.csite.2020.100681>
 - a. The effectiveness of two-dimensional heliostat tracking in solar thermal tower systems is evaluated through a series of case studies. The results demonstrate that dynamic tracking significantly increases solar energy concentration and thermal output compared to static configurations. By continuously adjusting to the sun's position, the system achieves improved efficiency and energy capture across varying conditions. These findings reinforce the importance of adaptive solar alignment in energy systems. Even in smaller-scale applications, optimizing the angle and orientation of solar panels can lead to measurable performance gains. The study provides practical insight into how tracking mechanisms can enhance energy efficiency in both large and compact solar technologies.
- 8. **Kallayil, A., et al. (2025).** Adaptive Smart Materials in Architecture: Enhancing Durability and Sustainability in Modern Construction. *ACS Omega*. <https://doi.org/10.1021/acsomega.4c04943>
 - a. Adaptive smart materials are examined for their ability to respond dynamically to environmental conditions, improving durability and sustainability in modern construction. These materials can adjust their physical or thermal properties in response to external stimuli, allowing structures to maintain performance under changing conditions. The study highlights advancements in material science that enable energy-efficient and resilient design. The application of such materials extends beyond architecture into wearable and portable technologies. Incorporating adaptive materials into compact systems could enhance performance by responding to temperature fluctuations or environmental stress. This research supports the integration of smart materials into innovative engineering designs aimed at improving efficiency and longevity.
- 9. **Li, Y. M., et al. (2022).** Tailored modular assembly derived self-healing polythioureas with largely tunable properties covering plastics, elastomers and fibers. *Nature Communications*, 13(1). <https://doi.org/10.1038/s41467-022-30364-x>
 - a. Self-healing polymer materials are introduced with tunable properties suitable for a wide range of applications, including plastics, elastomers, and fibers. The study demonstrates how these materials can repair structural damage autonomously, maintaining functionality

over time. The modular design approach allows for customization of mechanical and thermal properties, expanding potential use cases.

- b. Such material innovations are particularly valuable for products exposed to repeated mechanical stress. In wearable devices or backpacks, self-healing materials could significantly extend product lifespan and reliability. The research highlights the potential for integrating advanced materials into engineering designs to improve durability and reduce maintenance.
10. **Newsire, P. (2024).** BLUETTI Introduces Handsfree Solar Backpack Power Stations for Non-Stop Charging on the Go. *PR Newswire US*.
 - a. Recent developments in consumer technology demonstrate the growing adoption of solar-powered wearable energy systems. The introduction of hands-free solar backpack power stations reflects a shift toward integrating renewable energy into everyday products. These systems are designed to provide continuous charging capabilities while maintaining portability and convenience for users. This source provides real-world context for the application of solar-powered designs. It illustrates how concepts explored in academic research are being translated into commercially viable products. The information supports the relevance of ongoing innovation in portable renewable energy technologies.
11. **Riad, A., et al. (2020).** Bio-sun tracker engineering self-driven by thermo-mechanical actuator for photovoltaic solar systems. *Case Studies in Thermal Engineering, 21*. <https://doi.org/10.1016/j.csite.2020.100709>
 - a. A bio-inspired solar tracking system driven by thermo-mechanical actuators is presented as an energy-efficient alternative to traditional tracking mechanisms. The system adjusts its position based on temperature changes, eliminating the need for external power input. This passive tracking approach enhances photovoltaic performance while maintaining simplicity and sustainability.
 - b. Such designs are particularly advantageous for low-power or portable systems. Incorporating self-driven tracking mechanisms can improve energy capture without increasing system complexity or energy consumption. The study highlights innovative approaches to enhancing solar efficiency through passive design principles.
12. **Sriomreun, P. (2020).** An optimum design for air filter housing-louver: Enhancing filter life-time. *Case Studies in Thermal Engineering, 21*. <https://doi.org/10.1016/j.csite.2020.100701>
 - a. Optimization of air filter housing design is explored with the goal of improving airflow and extending filter lifespan. Structural modifications to the louver system result in more efficient air distribution and reduced clogging. The study demonstrates that relatively simple design changes can lead to significant improvements in performance and maintenance requirements.
 - b. Efficient airflow management is essential in many engineering applications, particularly those involving cooling or ventilation. For compact systems, improved airflow can

enhance heat dissipation and overall efficiency. This research reinforces the importance of design optimization in achieving long-term system performance.

13. **Sultan, S. M., et al. (2020).** A new method for reducing the performance evaluation cost of the photovoltaic module cooling techniques using the photovoltaic efficiency difference factor. *Case Studies in Thermal Engineering*, 21. <https://doi.org/10.1016/j.csite.2020.100682>
 - a. A simplified method for evaluating photovoltaic cooling techniques is introduced, reducing the cost and complexity of performance analysis. The proposed efficiency difference factor allows for quicker comparisons between cooling strategies without requiring extensive experimental setups. This approach enables more efficient decision-making in the design and testing of solar systems.
 - b. For smaller-scale projects, reducing evaluation time and resource requirements is especially beneficial. The method provides a practical tool for selecting optimal cooling solutions in portable or wearable solar technologies. This work supports more streamlined and cost-effective design processes.
14. **Taverne, J., et al. (2018).** Design of solar powered charging backpack. *International Journal of Power Electronics and Drive Systems*, 9(2), 848–858.
<https://doi.org/10.11591/IJPEDS.V9.I2.PP848-858>
 - a. A detailed design and evaluation of a solar-powered charging backpack demonstrates the feasibility of portable renewable energy systems. The integration of photovoltaic panels, energy storage, and power regulation components enables the system to charge small electronic devices effectively. Performance testing under real-world conditions confirms its practicality, while also addressing considerations such as weight distribution, durability, and user convenience.
 - b. This study serves as a direct reference for similar engineering projects, providing both technical guidance and practical insight into system integration. It highlights the importance of balancing efficiency with usability, ensuring that the final design meets both performance and user needs. The work reinforces the potential of solar-powered wearables as viable solutions for mobile energy access.

Citations

1. Abdelmalek, Z., Rehman, K. U., Al-Mdallal, Q. M., Al-Kouz, W., & Malik, M. Y. (2020). Dynamics of thermally magnetized grooved flow field having uniformly heated circular cylinder: Finite element analysis. *Case Studies in Thermal Engineering*, 21. <https://doi.org/10.1016/j.csite.2020.100718>
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